

Unit 4.4 L'Hospital's Rule

Note Title

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$\frac{\infty}{\infty} = 1$? $\left(\frac{\infty}{\infty}\right)$ A form, not a number

$$\lim_{x \rightarrow \infty} \frac{x}{1+x^2} = 0$$

$$\lim_{x \rightarrow \infty} \frac{3x+1}{4x+2} = \frac{3}{4}$$

$$\lim_{x \rightarrow \infty} \frac{1+x^2}{x} = \infty$$

Could be anything!

There are 7 indeterminate forms

$$\left(\frac{0}{0}\right), \left(\frac{\pm \infty}{\pm \infty}\right), (\infty - \infty), (0 \times \infty)$$

$$0^0, \infty^0 \text{ and } 1^\infty$$

Note: $(\infty \times \infty) \rightarrow \infty$ $(\infty + \infty) \rightarrow \infty$

$\infty \leftarrow \left(\pm \frac{k}{\infty}\right)$ $\left(\frac{\pm \infty}{k}\right) \rightarrow \infty$ $(\infty^\infty) \rightarrow \infty$

are NOT indeterminate forms.

NB.

Working with indeterminate forms

L'Hospital's rule: Summary

If $\lim_{x \rightarrow *} \frac{f(x)}{g(x)}$ is of the form $\left(\frac{0}{0}\right)$ or $\left(\frac{\pm\infty}{\pm\infty}\right)$

where $\underline{*} = \infty, -\infty$ or (a) — any number

$$\text{then } \lim_{x \rightarrow *} \frac{f(x)}{g(x)} = \lim_{x \rightarrow *} \frac{f'(x)}{g'(x)}$$

Examples:

Forms: $\left(\frac{0}{0}\right), \left(\frac{\pm\infty}{\pm\infty}\right)$

$$1. \lim_{x \rightarrow \infty} \frac{x}{1+x^2} = \lim_{x \rightarrow \infty} \frac{1}{2x} = 0 \quad \left(\frac{\infty}{\infty}\right)$$

$$2. \lim_{x \rightarrow \infty} \frac{3^x}{x^2} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow \infty} \frac{3^x \cdot \ln 3}{2x} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow \infty} \frac{3^x \cdot \ln 3 \cdot \ln 3}{2} = \infty$$

$$3. \lim_{x \rightarrow -\infty} \frac{3x}{x^2} \quad \left(\frac{-\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow -\infty} \frac{3}{2x} = 0$$

$$4. \quad \lim_{x \rightarrow \infty} \frac{\ln x}{x^2} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{2x} \quad \left(\frac{\infty}{\infty} \right) \\ = \lim_{x \rightarrow \infty} \frac{1}{2x^2} = 0.$$

$$5. \quad \lim_{x \rightarrow -\infty} \frac{\ln x}{x^2} \text{ not defined.}$$

$$6. \quad \lim_{x \rightarrow 0} \frac{2x}{e^{3x} - 1} = \lim_{x \rightarrow 0} \frac{2}{3e^{3x}} \quad \left(\frac{0}{0} \right) \\ = \frac{2}{3}.$$

$$7. \quad \lim_{x \rightarrow \pi/2} \frac{\cos x}{2x - \pi} \quad \left(\frac{0}{0} \right) \\ = \lim_{x \rightarrow \pi/2} \frac{-\sin x}{2} = -\frac{1}{2}.$$

$$8. \quad \lim_{x \rightarrow \infty} \frac{x \ln x}{x + \ln x} \quad \left(\frac{\infty}{\infty} \right) \\ = \lim_{x \rightarrow \infty} \frac{\ln x + \cancel{x \cdot \frac{1}{x}}}{1 + \frac{1}{x}} = 1. \quad \left(\frac{\infty}{1} \right) \\ = \infty.$$

$$9. \quad \lim_{x \rightarrow \infty} \frac{x^2}{4 + e^x} \quad \left(\frac{\infty}{\infty} \right) \\ = \lim_{x \rightarrow \infty} \frac{2x}{e^x} \quad \left(\frac{\infty}{\infty} \right) \\ = \lim_{x \rightarrow \infty} \frac{2}{e^x} = 0. \quad \left(\frac{0}{\infty} \right)$$

Examples Form: $(\infty - \infty)$

- Factorize or
- Add fractions.

$$\begin{aligned} 1. \quad & \lim_{x \rightarrow \infty} (x^3 - x^4) \\ &= \lim_{x \rightarrow \infty} x^3 (1 - x) = -\infty \end{aligned}$$

$$\begin{aligned} & (\infty - \infty) \\ & (\infty \cdot (-\infty)) \end{aligned}$$

$$\begin{aligned} 2. \quad & \lim_{x \rightarrow -\infty} (x^4 + x^3) \\ &= \lim_{x \rightarrow -\infty} x^3 (x + 1) = \infty \end{aligned}$$

$$\begin{aligned} & (\infty - \infty) \\ & (\infty \cdot \infty) \end{aligned}$$

$$\begin{aligned} 3. \quad & \lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right) \\ &= \lim_{x \rightarrow 1^+} \frac{x-1 - \ln x}{\ln x (x-1)} \end{aligned}$$

WB.

$$\begin{aligned} & (\infty - \infty) \\ & \left(\frac{0}{0} \right) \end{aligned}$$

$$= \lim_{x \rightarrow 1^+} \frac{1 - \frac{1}{x}}{\frac{1}{x}(x-1) + \ln x}$$

$$= \lim_{x \rightarrow 1^+} \frac{1 - \frac{1}{x}}{1 - \frac{1}{x} + \ln x}$$

$$\left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1^+} \frac{\frac{1}{x^2}}{\frac{1}{x^2} + \frac{1}{x}}$$

$$= \frac{1}{2}$$

~~→~~

Examples Form $(\infty \times 0)$
• Shape into a fractional function

1. $\lim_{x \rightarrow \infty} x e^{-x}$ $(\infty \times 0)$

$= \lim_{x \rightarrow \infty} \frac{x}{e^x}$ $(\frac{\infty}{\infty})$

$= \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$

2. $\lim_{x \rightarrow \infty} x e^{\frac{3}{x}}$ $(\infty \times 1)$

$= \infty$ Not indeterminate.

3. $\lim_{x \rightarrow \infty} \frac{1}{x} \ln x$ $(0 \times \infty)$

$= \lim_{x \rightarrow \infty} \frac{\ln x}{x}$ $(\frac{\infty}{\infty})$

$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0$

Examples Forms (0^0) , (∞^0) , (1^∞)

1. $\lim_{x \rightarrow \infty} \left(1 + \frac{4}{x}\right)^{3x}$

Let $y = \left(1 + \frac{4}{x}\right)^{3x}$

$\Rightarrow \ln y = 3x \ln \left(1 + \frac{4}{x}\right)$

$$\Rightarrow \lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} 3x \ln\left(1 + \frac{4}{x}\right) \quad (\infty \times 0)$$

$$= \lim_{x \rightarrow \infty} 3 \cdot \frac{\ln\left(1 + \frac{4}{x}\right)}{\frac{1}{x}} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x \rightarrow \infty} 3 \cdot \frac{1}{1 + \frac{4}{x}} \times \frac{-4}{x^2}$$

$$= 12$$

$\uparrow$
 $\ln \text{ const.}$
 $\downarrow$

$$\Rightarrow \ln\left(\lim_{x \rightarrow \infty} y\right) = 12$$

NB Do not differentiate!

$$\Rightarrow \lim_{x \rightarrow \infty} y = e^{12}$$

2. $\lim_{x \rightarrow \infty} (e^x + bx)^{2/x}$

Let $y = (e^x + bx)^{2/x}$

$$\Rightarrow \ln y = \frac{2}{x} \ln(e^x + bx)$$

$$\Rightarrow \lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{2}{x} \ln(e^x + bx) \quad (0 \times \infty)$$

$$= \lim_{x \rightarrow \infty} 2 \cdot \frac{\ln(e^x + bx)}{x} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow \infty} 2 \cdot \frac{\frac{e^x + b}{e^x + bx}}{1} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow \infty} 2 \cdot \frac{e^x}{e^x + b} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow \infty} 2 \cdot \frac{e^x}{e^x}$$

$$\Rightarrow \lim_{x \rightarrow \infty} y = e^2$$

$$3. \lim_{x \rightarrow 0^+} (2x)^{1/\ln x}$$

$$\text{Let } y = (2x)^{\frac{1}{\ln x}}$$

$$\Rightarrow \ln y = \frac{1}{\ln x} \cdot \ln 2x \quad (0 \cdot \infty)$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+}$$

$$= \lim_{x \rightarrow 0^+}$$

$$= \lim_{x \rightarrow 0^+}$$

$$\frac{\ln 2x}{\ln x}$$

$$\frac{\cancel{\frac{1}{x}}}{\cancel{\frac{1}{x}}} = 1$$

$$\Rightarrow \lim_{x \rightarrow 0^+} y = e^1$$

$$\left| \frac{d}{dx} C \cdot f(x) \right| = C \cdot \frac{d}{dx} f(x)$$

$$\frac{\ln x}{\frac{2}{2x} \cdot \frac{1}{x}}$$